

Class 2 – Meningioma Tumor: Patient & Caregiver Guide

An educational guide for people whose brain MRI suggests a meningioma — a tumor that grows from the meninges, the protective membranes covering the brain and spinal cord. It explains what a meningioma is, its grades, symptoms, diagnosis, MRI features, treatment, recovery, nutrition, rehabilitation, warning signs, outlook, and answers to common questions.

Educational disclaimer: This document is for general education and for use in a retrieval-augmented (RAG) AI assistant that supports patients after an MRI-based brain scan. It is NOT a medical diagnosis and does NOT replace consultation with qualified healthcare professionals such as a neurologist, neurosurgeon, or oncologist. Every person is different, and only your own medical team can confirm a diagnosis and plan treatment. If you have severe or sudden symptoms, seek emergency medical care immediately.

1. Overview

A meningioma is a tumor that grows from the meninges — the thin layers of tissue that cover and protect the brain and spinal cord. It is the most common primary brain tumor in adults. Importantly, a meningioma grows outside the brain itself and presses on it, rather than growing inside the brain tissue.

Are they benign or malignant? The great majority of meningiomas are benign (non-cancerous, WHO Grade I), slow-growing, and do not spread. A smaller number are atypical (Grade II) or, rarely, malignant (Grade III), which grow faster and are more likely to return.

Where they occur: along the surface of the brain, at the skull base, near the front of the brain (falx and convexity), and around the spinal cord. Epidemiology: meningiomas are common and are often found by chance on scans done for other reasons. Age groups: most common in middle-aged and older adults. Gender: about twice as common in women, which suggests a hormonal influence.

Disease progression and prognosis: most meningiomas grow very slowly, and many cause no symptoms for years. The outlook is generally very good, especially for benign tumors that can be fully removed or safely monitored. Higher-grade meningiomas need closer follow-up.

2. Brain Anatomy

Knowing the structures around a meningioma explains its symptoms and treatment.

- **Meninges:** three protective layers (dura, arachnoid, pia) that wrap the brain. Meningiomas usually arise from the arachnoid cells of these layers.
- **Brain surface (cortex):** the meningioma presses on nearby cortex, which may control movement, sensation, vision, or thinking depending on location.
- **Skull base:** many meningiomas sit here, close to important cranial nerves.
- **Cranial nerves:** nerves for smell, vision, eye movement, facial sensation, and hearing can be pressed by nearby meningiomas, causing specific symptoms.
- **Blood vessels and venous sinuses:** large veins (sinuses) run in the dura; meningiomas near them can be more complex to remove.
- **Optic nerves and chiasm:** meningiomas near the front of the skull base can press on the vision pathways.

Relationship to surrounding tissue: because a meningioma grows outside the brain and pushes on it, surgeons can often separate it from the brain, and complete removal is more achievable than with gliomas. However, tumors wrapped around nerves, vessels, or venous sinuses can be harder to remove completely.

3. What Causes a Meningioma?

The exact cause is usually unknown, but several factors are linked to higher risk. A meningioma is not caused by anything the patient did wrong and is not contagious.

Established Risk Factors

- Previous radiation to the head: the strongest known risk factor, sometimes decades later.
- Age: risk rises with age.
- Female sex: meningiomas are about twice as common in women.
- Neurofibromatosis type 2 (NF2): an inherited condition that raises the risk of meningiomas.

Possible Risk Factors

- Hormonal influences: meningiomas can carry hormone receptors and may grow during pregnancy or with hormone therapy; the exact role is still studied.
- Obesity has been linked to slightly higher risk in some studies.

Protective / Not Proven Factors

- No specific action is proven to prevent meningiomas.
- A healthy lifestyle supports overall brain health but is not proven to prevent these tumors.

4. Types and Classification (WHO Grades)

Meningiomas are graded I to III. Most are Grade I (benign). The grade guides treatment and follow-up.

WHO Grade	Growth & severity	Recurrence	Typical treatment	General prognosis
Grade I	Benign, slow-growing; the great majority of meningiomas.	Low, especially if fully removed	Observation or surgery; sometimes radiation	Very good
Grade II	Atypical; grows faster and is more likely to return.	Moderate	Surgery, often followed by radiation	Good but needs closer follow-up
Grade III	Malignant (anaplastic); rare, aggressive.	High	Surgery plus radiation; close monitoring	More guarded

Note: there is no Grade IV for meningiomas. Even benign (Grade I) meningiomas can cause problems if they press on vital structures, so location matters as much as grade.

5. Common Symptoms

Many meningiomas cause no symptoms and are found by chance. When symptoms occur, they come from the tumor pressing on nearby brain or nerves, and depend on its location.

- Headache: from slow pressure build-up; often mild and gradual.
- Seizures: when the tumor irritates the nearby brain surface.
- Weakness or numbness: on one side, if the movement or sensation area is pressed.
- Vision changes: blurred vision or loss of part of the visual field from tumors near the optic pathways.
- Hearing changes: from meningiomas near the hearing nerve at the skull base.
- Memory problems and difficulty concentrating: from frontal-area pressure.
- Personality changes: with tumors at the front of the brain.
- Balance problems: from tumors near the back of the brain or skull base.
- Smell changes: from tumors at the front of the skull base.
- Fatigue: general tiredness, common with brain tumors and treatment.

When to get emergency care: a new seizure, sudden severe headache, sudden weakness, sudden vision loss, repeated vomiting, or a drop in consciousness are emergencies — seek immediate medical help.

6. Diagnosis

Meningiomas are often diagnosed confidently on imaging because they have a typical appearance. A biopsy is not always needed before surgery.

- Medical history: symptoms, their timing, past radiation, and family history.
- Neurological examination: checks vision, hearing, strength, sensation, balance, and thinking.
- MRI: the main test; shows the tumor, its location, and its relationship to the brain and nerves without radiation.
- Contrast MRI: meningiomas typically enhance strongly and evenly with dye, and often show a 'dural tail' (a tail of thickened membrane) that is very characteristic.
- CT scan: shows bone changes and calcification well, and is useful in emergencies.
- PET scan: occasionally used to assess activity or distinguish tumor from other tissue.
- MR spectroscopy: may help in unclear cases.
- Biopsy: only some cases need one before treatment; often the diagnosis is confirmed on the tissue removed at surgery.
- Blood tests: to check general health before treatment.

Advantages and limitations: MRI with contrast is usually enough to strongly suggest a meningioma, but the final grade is confirmed by examining the removed tissue.

7. MRI Characteristics

Meningiomas have a classic MRI appearance that helps distinguish them from other tumors.

- Location: attached to the meninges on the surface of the brain or at the skull base, outside the brain tissue.

- Shape: usually round or oval with smooth, well-defined edges.
- Size: variable; can be small and incidental or large.
- Contrast enhancement: strong, uniform enhancement after dye.
- Dural tail: a characteristic tail of enhancing thickened dura next to the tumor.
- Edema: some surrounding brain swelling may occur, especially with larger tumors.
- Mass effect: the tumor pushes on nearby brain but does not invade it in most cases.
- Calcification: common and well seen on CT.
- Necrosis: uncommon in benign meningiomas; more possible in higher grades.

A typical meningioma looks like a smooth, round, strongly and evenly enhancing mass on the brain's surface with a dural tail — features that make it fairly recognisable to radiologists.

8. Treatment Options

Treatment depends on the tumor's size, location, grade, symptoms, and your health. Many small, symptom-free meningiomas are simply watched.

Observation (Watchful Waiting)

Small meningiomas that cause no symptoms — often found by chance — are frequently monitored with regular MRIs. Because they grow so slowly, many never need treatment, especially in older adults.

Surgery

- Craniotomy: the main treatment for symptomatic or growing tumors; a section of skull is temporarily removed to take out the tumor.
- Endoscopic surgery: used for some skull-base meningiomas through natural openings.
- Complete removal: often achievable because the tumor sits outside the brain; this can be curative for benign meningiomas.
- Partial removal: when the tumor wraps around nerves, vessels, or venous sinuses, surgeons remove what is safe and may add radiation.
- Benefits: relieves pressure, can cure benign tumors, and confirms the grade.
- Risks: bleeding, infection, swelling, seizures, and possible nerve injury depending on location.
- Recovery: hospital stay of a few days, with improvement over weeks.

Radiation Therapy

- Stereotactic radiosurgery (Gamma Knife, CyberKnife): highly focused radiation, good for small tumors or those left after surgery, often in one or a few sessions.
- External beam radiation: fractionated radiation over several weeks for larger or higher-grade tumors.
- Side effects: tiredness, local hair loss, and sometimes temporary swelling.
- Recovery: fatigue eases over weeks after treatment.

Chemotherapy

Chemotherapy is not a standard treatment for most meningiomas and is used mainly for rare, aggressive tumors that have not responded to surgery and radiation, usually within specialist care or trials.

Targeted Therapy and Immunotherapy

These are not standard for meningiomas but are being studied for rare, difficult-to-treat cases, mainly in clinical trials.

Hormone Therapy

Because some meningiomas carry hormone receptors, hormone-related treatments have been studied, but hormone therapy is not a standard treatment. Your doctor may review any hormone medicines you take, especially during pregnancy planning.

9. Recovery Timeline

Recovery is usually smooth for benign meningiomas, especially after complete removal. These are general guides.

Pathway	What to expect
After surgery	Hospital stay usually 2–5 days; most fatigue settles over 4–6 weeks; full recovery over 2–3 months.
After radiation/radiosurgery	Radiosurgery is often outpatient; fatigue is mild and settles over a few weeks.
Observation	No treatment recovery; regular MRIs monitor the tumor.
Higher-grade tumors	May need combined surgery and radiation, with a longer recovery and closer follow-up.

- Return to work: often a few weeks after surgery for desk jobs; longer for physically demanding roles.
- Return to school: gradual return once recovered and cleared.
- Driving: usually paused after surgery or a seizure; resume after medical and licensing clearance.
- Physical activity: gentle activity early; avoid heavy lifting and contact activity until cleared.

Expected milestones: relief of pressure symptoms, steady return of energy over weeks, and good long-term function for most benign tumors.

10. Home Care

Home care supports healing and safety between visits.

- Medication adherence: take any anti-seizure medicines, steroids, or pain relief exactly as prescribed.
- Sleep: keep regular, adequate sleep to aid recovery.
- Hydration: drink enough water each day.
- Stress reduction: use relaxation and gentle routines.

- Activity restrictions: avoid driving until cleared, and avoid heavy lifting and risky activities early after surgery.
- Rest and pacing: balance activity with rest; fatigue is normal for a few weeks.
- Mental health: seek support for worry or low mood.
- Follow-up appointments: keep all scans and reviews; note any new symptoms.
- Wound care: keep the surgical wound clean and dry as instructed, and watch for signs of infection.

11. Nutrition

Good nutrition supports recovery from surgery or radiation and general brain health, helps the body cope with treatment, and keeps energy and mood steady. There is no single 'anti-tumor' diet, but a balanced, mostly whole-food diet is widely recommended. Always follow any specific advice from your medical team or dietitian, especially during treatment.

Key Nutrients and Why They Help

Nutrient	Why it helps	Vegetarian sources	Non-vegetarian sources
Protein	Repairs tissue and supports healing after surgery or treatment.	Dal, beans, chickpeas, rajma, soya, paneer, tofu, nuts	Eggs, chicken, fish, lean meat
Omega-3 fats	Support brain-cell health and reduce inflammation.	Flaxseed, chia, walnuts, soybean oil	Fatty fish (salmon, mackerel, sardines)
Vitamin D	Supports bone and immune health; many patients are low.	Sunlight, fortified milk/cereals, mushrooms	Egg yolk, fatty fish
Vitamin B12	Needed for healthy nerves and blood.	Fortified foods, dairy, supplements if vegan	Eggs, dairy, fish, meat
Vitamin C	Helps wound healing and immunity.	Amla, orange, guava, lemon, capsicum, tomato	(mainly plant foods)
Vitamin E	An antioxidant that protects cells.	Nuts, seeds, vegetable oils, spinach	(mainly plant foods)
Antioxidants	Protect cells from damage; from colourful produce.	Berries, leafy greens, carrots, beetroot, tomatoes	(mainly plant foods)
Healthy fats	Provide steady energy and support the brain.	Olive oil, mustard oil, nuts, avocado, seeds	Fish
Whole grains	Give steady energy and fibre for digestion.	Oats, brown rice, millets (jowar, bajra, ragi), whole wheat	(mainly plant foods)
Hydration (water)	Prevents tiredness, headaches, and constipation.	Water, buttermilk, coconut water, soups	Water, broths

Sample Daily Meal Suggestions

- Breakfast: vegetable oats or besan chilla or eggs with whole-wheat toast and a fruit.

- Mid-morning: a fruit (apple, guava, orange) or a handful of nuts.
- Lunch: 2 multigrain rotis, dal or beans, a large bowl of vegetables, curd, and salad.
- Evening: buttermilk, sprouts, or green tea with roasted chana.
- Dinner: brown rice or roti with vegetables and a protein (fish, chicken, tofu, or dal).
- Through the day: sip water regularly to stay well hydrated.

During chemotherapy or if you have swallowing, taste, or appetite problems, ask your team for tailored advice. Do not start high-dose supplements on your own, as some can interfere with treatment.

12. Foods to Limit

Some foods and habits do not directly cause brain tumors but can slow recovery, worsen general health, or interact with treatment. It is wise to limit the following.

- Alcohol: can interact with medicines (including anti-seizure and chemotherapy drugs), affect the brain, and disturb sleep. Best avoided or strictly limited.
- Smoking and tobacco: harm blood vessels and overall health and slow healing. Stopping is strongly recommended.
- Highly processed foods: often high in salt, sugar, and unhealthy fat and low in nutrients.
- Sugary beverages: sodas and sweetened juices add empty calories and can worsen energy swings.
- Excess salt: too much salt can raise blood pressure and cause fluid retention.
- Excess caffeine: large amounts of coffee, tea, or energy drinks can worsen anxiety, tremor, and sleep problems.
- Trans fats: found in some fried and bakery items; bad for heart and blood vessels.
- Ultra-processed foods: packaged snacks and ready meals; keep these occasional.

In short, a fresh, home-cooked, balanced diet with less salt, sugar, alcohol, and no smoking supports recovery and overall brain health.

13. Rehabilitation

Many people need little or no rehabilitation after benign meningioma surgery. When the tumor or surgery affects specific functions, targeted therapy helps recovery.

- Physical therapy: restores strength, walking, and balance if these were affected.
- Occupational therapy: helps with daily tasks and returning to independence.
- Speech and language therapy: for speech or swallowing difficulties.
- Vision therapy: supports adaptation if vision is affected.
- Cognitive rehabilitation: exercises for memory and concentration.
- Balance training: reduces falls, useful after skull-base surgery.
- Memory exercises: strategies and reminders to support daily life.

Duration and goals: rehabilitation may last a few weeks to months if needed. The goal is to regain independence and quality of life. Outcomes are generally good, especially for benign tumors.

14. Lifestyle Recommendations

Healthy daily habits support brain health, mood, and recovery. Adapt these to your abilities and your doctor's advice.

- Regular exercise: gentle activity such as walking, as approved by your doctor, improves energy, mood, and sleep.
- Stress management: reduce stress with breathing exercises, hobbies, and pacing your day.
- Meditation: mindfulness or guided relaxation can ease anxiety and improve focus.
- Yoga: gentle yoga (with medical clearance) supports flexibility, balance, and calm.
- Sleep hygiene: keep a regular sleep schedule, a dark quiet room, and limit screens before bed; aim for 7–9 hours.
- Hydration: drink water through the day to avoid tiredness and headaches.
- Weight management: keep a healthy weight with balanced food and activity.
- Mental wellness: seek support for low mood or anxiety; counselling can help.
- Family support: involve family in appointments, medication, and daily routines.
- Social engagement: stay connected with friends and community to protect mood and thinking skills.

15. Possible Complications

Complications are uncommon for benign meningiomas but can occur, from the tumor or its treatment.

- Brain swelling (edema): can cause headaches and pressure; treated with steroids.
- Hydrocephalus: fluid build-up, occasionally needing a drain or shunt.
- Seizures: may occur before or after surgery; managed with medicines.
- Vision or hearing loss: from tumors pressing on those nerves, sometimes lasting.
- Hormonal imbalance: rare, mainly with skull-base tumors near hormonal structures.
- Neurological deficits: weakness, numbness, or facial changes from nerve involvement.
- Recurrence: possible, higher for atypical and malignant grades, so follow-up continues.
- Infection: a small risk after surgery.
- Blood clots: reduced mobility raises risk; report leg swelling or breathlessness.
- Mood disorders: anxiety and low mood are common and treatable.

16. Emergency Warning Signs

Seek emergency medical help right away for any of these:

- A sudden, severe ('worst ever') headache.
- A new seizure or a seizure lasting more than 5 minutes.
- Loss of consciousness or sudden severe drowsiness.
- Repeated vomiting with headache.

- Sudden confusion or difficulty speaking.
- Sudden weakness or paralysis, especially on one side.
- Sudden vision loss or double vision.
- Rapidly worsening symptoms of any kind.

These can signal raised pressure in the brain or bleeding. Do not wait — call emergency services or go to the nearest hospital.

17. Follow-Up Schedule

Follow-up watches for regrowth and manages any effects. The schedule depends on the grade and whether the tumor was removed or is being observed.

- Initial follow-up: within a few weeks of surgery to check healing and results.
- MRI scans: a baseline scan after surgery, then periodic scans (often yearly for benign tumors, more often for higher grades or when observing).
- Neurological examination: at each visit to track vision, strength, and thinking.
- Blood tests: as needed, especially if other treatment is used.
- Long-term monitoring: continues for years because meningiomas can regrow slowly.
- Recurrence surveillance: new or returning symptoms prompt an earlier scan.

18. Long-Term Outlook

The outlook for most meningiomas is very good, particularly for benign (Grade I) tumors.

- Prognosis: excellent for fully removed benign meningiomas; many people are effectively cured.
- Quality of life: usually very good, with most people returning to normal activities.
- Recurrence risk: low for benign tumors, higher for atypical and malignant grades, so monitoring continues.
- Living with the condition: many people simply have periodic scans if the tumor is small and stable.
- Psychological support: counselling helps with the worry of monitoring or diagnosis.

Your own specialists can give the most accurate outlook based on the tumor's grade, size, location, and how much was removed.

19. Prevention and Risk Reduction

There is no proven way to prevent meningiomas, but general steps support brain health.

- Healthy lifestyle: balanced diet, regular activity, healthy weight, good sleep, and not smoking.
- Avoid unnecessary radiation exposure: since past head radiation is the main known risk, only have imaging that is clinically needed.
- Manage chronic conditions: control blood pressure, weight, and other health issues.
- Regular medical check-ups: report new headaches, seizures, or vision changes promptly.
- Screening: there is no routine screening for meningiomas; scans are done when symptoms or risk factors are present. People with NF2 may have regular monitoring.

- What cannot be prevented: most meningiomas arise without an avoidable cause.

20. Frequently Asked Questions (FAQ)

Below are common questions from patients and caregivers about meningiomas. Answers are general; your own medical team can advise for your situation.

Q1. What is a meningioma?

A meningioma is a tumor that grows from the meninges, the protective membranes covering the brain. It grows outside the brain and presses on it rather than growing inside it.

Q2. Is a meningioma cancer?

Usually not. Most meningiomas are benign (non-cancerous) and slow-growing. A small number are atypical or malignant and behave more aggressively.

Q3. Can a meningioma spread?

Benign meningiomas do not spread to other organs. They can grow larger and press on nearby structures. Rare malignant ones can, uncommonly, spread.

Q4. Will I need surgery?

Not always. Small, symptom-free meningiomas are often just monitored. Surgery is used when the tumor causes symptoms, is growing, or is large.

Q5. Can a meningioma come back after removal?

Benign tumors that are fully removed rarely return, but recurrence is possible, especially for higher grades, so follow-up scans continue.

Q6. Is a meningioma hereditary?

Usually not. Most occur by chance. The inherited condition neurofibromatosis type 2 (NF2) does raise the risk.

Q7. Why are meningiomas more common in women?

They occur about twice as often in women and can carry hormone receptors, suggesting a hormonal influence, though the exact reason is not fully understood.

Q8. Can I drive?

Driving is usually paused after surgery or a seizure. You can often return after clearance from your doctor and the licensing authority, which varies by region.

Q9. Can I travel?

Yes, once stable and cleared by your doctor. Avoid travel immediately after surgery and carry a medical summary and your medicines.

Q10. Can I fly?

Flying is generally safe once cleared, but avoid it soon after brain surgery until your doctor agrees.

Q11. Can I work?

Most people return to work, often within a few weeks after surgery for desk jobs. Physically demanding jobs may need longer.

Q12. Can I exercise?

Gentle activity is encouraged during recovery. Avoid heavy lifting and contact activity until your doctor clears you.

Q13. Can I have children / is pregnancy safe?

Many women do. Because some meningiomas can grow during pregnancy, discuss timing and monitoring with your doctor, especially if you have a known tumor.

Q14. Can I drink alcohol?

Limit alcohol, especially during recovery or if you take anti-seizure medicines, as it can interact with them and affect the brain.

Q15. Is an MRI safe?

Yes. MRI uses magnets, not radiation. Tell staff about any metal implants or a pacemaker. Contrast dye is generally safe.

Q16. Will I lose my memory?

Most people do not. Memory or concentration can be affected by frontal tumors or surgery, and often improves; rehabilitation can help.

Q17. What symptoms require emergency care?

A new or prolonged seizure, sudden severe headache, sudden weakness, vision loss, repeated vomiting, or loss of consciousness — seek emergency care at once.

Q18. How often will I need follow-up scans?

Often a baseline scan after surgery, then usually yearly for benign tumors, and more often for higher grades or when observing a small tumor.

Q19. What does 'benign' mean here?

Benign means non-cancerous and slow-growing. However, even a benign meningioma can cause problems if it presses on important structures.

Q20. What is a 'dural tail' on my scan?

It is a characteristic tail of thickened membrane next to the tumor seen on MRI, which helps radiologists recognise a meningioma.

Q21. Can a meningioma be completely cured?

Often yes. A benign meningioma that is fully removed can be effectively cured, though follow-up scans continue for a time.

Q22. Do I need treatment if I have no symptoms?

Not always. Small, symptom-free tumors are commonly watched with scans, and many never need treatment, especially in older adults.

Q23. What is Gamma Knife or CyberKnife?

These are types of stereotactic radiosurgery — highly focused radiation that treats small tumors precisely, often in one or a few sessions, without an incision.

Q24. Will radiation make me sick?

Radiosurgery is usually well tolerated; the main effect is mild tiredness. Larger course radiation can cause fatigue and local hair loss.

Q25. Is chemotherapy used for meningiomas?

Rarely. It is reserved for uncommon aggressive tumors that do not respond to surgery and radiation, usually within specialist care.

Q26. Why do I have seizures?

A meningioma can irritate the nearby brain surface and trigger seizures. Anti-seizure medicines usually control them.

Q27. Is a meningioma contagious?

No. You cannot catch or pass on a brain tumor.

Q28. Will I need rehabilitation?

Many people need little or none. If movement, speech, or thinking is affected, targeted therapy helps recovery.

Q29. How is a meningioma diagnosed?

Usually confidently on MRI with contrast because of its typical appearance; the grade is confirmed by examining tissue removed at surgery.

Q30. Can I take supplements or hormones?

Check with your doctor. Because some meningiomas have hormone receptors, review any hormone medicines. Avoid unproven remedies during treatment.

Q31. Will my personality change?

Front-of-brain meningiomas can affect mood or behaviour; this often improves after treatment. Support and counselling help.

Q32. Can children get meningiomas?

They are much less common in children and, when they occur, may be linked to conditions such as NF2 or previous radiation.

Q33. How long is the hospital stay after surgery?

Often about 2–5 days for a craniotomy if recovery is smooth.

Q34. Is a meningioma painful?

The brain itself does not feel pain, but pressure can cause headaches. Pain is managed as part of your care.

Q35. Can I use my phone and computer normally?

Yes. Normal device use does not affect the tumor. Take breaks if screens worsen headaches.

Q36. What if the tumor is near a nerve or blood vessel?

Surgeons remove what is safe and may leave a small part to protect the nerve or vessel, sometimes adding radiation to the remainder.

Q37. Can a meningioma turn into cancer?

Most stay benign. A minority can be atypical or, rarely, become more aggressive, which is one reason follow-up scans are important.

Q38. How do I cope with the worry of monitoring?

Regular 'watch and wait' can be stressful. Counselling, support groups, and clear communication with your team help you cope.

Q39. Does diet cure a meningioma?

No diet cures it, but good nutrition supports recovery and general health. Be cautious of unproven 'cure' diets.

Q40. Who is on my care team?

Usually a neurosurgeon, radiation oncologist, radiologist, pathologist, neurologist, nurses, and rehabilitation therapists as needed.

Q41. What happens if it grows during observation?

If a watched tumor grows or causes symptoms, your team will discuss treatment, usually surgery or radiosurgery.

21. Metadata for Retrieval

Structured tags to help the AI assistant retrieve this document accurately.

Field	Value
Tumor Name	Meningioma
Alternative Names	Meningeal tumor, dural-based tumor
WHO Grade	WHO Grade I–III (most are Grade I, benign)
Tumor Category	Primary brain tumor of the meninges (extra-axial, outside the brain)
Brain Region	Meninges over the brain surface and skull base; sometimes spinal cord
Benign or Malignant	Usually benign; occasionally atypical or malignant
Typical Symptoms	Often none; headache, seizures, weakness, vision/hearing changes, memory and personality changes
Diagnostic Methods	Neurological exam, MRI, contrast MRI (dural tail), CT (calcification), sometimes biopsy
MRI Characteristics	Round, well-defined, strongly and uniformly enhancing extra-axial mass with dural tail; may calcify

Field	Value
Treatment Options	Observation, surgery (craniotomy), stereotactic radiosurgery/radiation; chemotherapy only in rare aggressive cases
Recovery Duration	Surgery 4–6 weeks initial, 2–3 months full; radiosurgery often outpatient
Follow-Up Frequency	Baseline MRI then usually yearly for benign; more often for higher grades
Common Complications	Edema, seizures, vision/hearing loss, neurological deficits, recurrence, hydrocephalus
Emergency Symptoms	New/prolonged seizure, sudden severe headache, paralysis, vision loss, loss of consciousness
Keywords	meningioma, benign brain tumor, meninges, dural tail, skull base tumor, craniotomy, Gamma Knife
Synonyms	Meningeal tumor, dural tumor, brain covering tumor
Related Medical Conditions	Neurofibromatosis type 2 (NF2), seizures, hydrocephalus, optic nerve compression